VICTORY 2027

ELECTRICITY AND MAGNETISM

LECTURE-05 PHYSICS

Shailendra sir



Topics To Be Covered

1. FUSE
2. SWITCHES
3. SOCKET
4. QUESTIONS


SHAILENDRA SIR
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Revision

Current Rating

$$P = VI$$

$$I = \frac{P}{V}$$



Current rating of a fuse

- In electric wiring for light and fan circuits a thin fuse wire of low current rating ($\approx 5A$) is used because the line wire has a current carrying capacity of 5 A.
- A thick fuse wire of higher current rating ($\approx 15A$) is used for heavy current consuming appliances such as air conditioner, geyser, washing machine, etc. because the line wire for such appliances has a current carrying capacity of 15 A.

Appliance	Power rating at 220 V (in watt)	Current drawn in running condition (in A)	Current rating for fuse
Electric bulb	60	0.27	1A fuse in line circuit
Television set	120	0.54	
Refrigerator	150	0.68	
Electric mixer	750	3.4 → 4A	
Room heater	1000	4.5	5A
Electric iron	1000	4.5	5A
Geyser	1500	7.0	8A
Electric kettle	2000	8.0	10A
Electric oven	3000	13.6	15A

$$\text{Current rating of fuse in a circuit} = \frac{\text{total power of appliances in circuit}}{\text{voltage of the supply}}$$



Current rating of a fuse

- A fuse rating should be slightly higher than the maximum safe current drawn by an appliance.
- Appliances with motors (mixer, water pump, etc.) draw a high starting current. Therefore, their fuse rating should be slightly higher than the starting current.
- Example: A 1.5 kW, 220 V water pump draws 10.4 A at starting and 6.8 A while running. Hence, a suitable fuse should have a rating of at least 10.5 A.

$P = V I$

$1500 = 220\ I$

$I = 6.81\ A$

M.C.B.

Magnetic Effect of electric Current

- Now a days instead of a fuse, a miniature circuit breaker (MCB) is used for each individual circuit.
- If due to short circuiting (or some fault), a heavy current flows in the line, the MCB falls down to switch off the circuit in a very short time (≈25 milli-second).
- After repairing the fault in the circuit, the MCB is then raised up.
- Thus the use of MCB is easier than a fuse because (i) it avoids the inconvenience of connecting a new fuse wire, and (ii) it is much safer due to its quick response.





QUESTIONS

Which is not the characteristic of a fuse wire ?

1. It has high resistance
2. It has low resistance
3. It is an alloy of lead and tin
4. It has low melting point



QUESTIONS

An electric motor of power 3 kW is to be operated at mains of 220 V. Find the current rating of the fuse to be connected with the motor.

$$P = V\,I$$

$$3000 = 220\ I$$

$$I = 13.63\ A$$

$$\downarrow$$

14 A or 15 A fuse

QUESTIONS

Imp

A house has main fuse of 5 A rating. 5 bulbs each of 60 W and 2 tube lights each of 40 W are used simultaneously. Find:

(i) the current drawn from the mains of 220 V, and

(ii) the number of additional bulbs each of 100 W which can also be lighted on a festival day?

Total Power $= 5 \times 60 + 2 \times 40$

$= 380$ W

$V = 220$ volts.

(i) $I = \frac{P}{V} = \frac{380}{220} = 1.72$ A.

(ii) Remaining Current $= 5 - 1.72$

$= 3.28$ A.

Let n bulbs are used

$P = V\ I$

$n \times 100 = 220 \times 3.28$

$n = 7.21$

≈ 7 Bulbs



QUESTIONS

The rating of a fuse connected in the lighting circuit is:

1. 15 A
2. 5 A
3. 10 A
4. zero

SWITCHES

A switch is an on-off device for current in a circuit (or in an appliance). It is connected in the live wire.

- **Single pole switch:** The switch used with an appliance to start or stop the flow of current in it, is the single pole switch. A single pole switch **disconnects only the live wire** from the appliance.



- **Double pole switch:** The main switch at the distribution board, used to switch on or off the mains, is the double pole switch. A double pole switch **disconnects both the live and neutral wires** simultaneously.

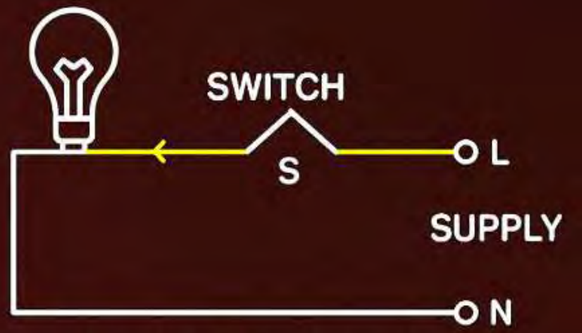
  The dual control switch used in a staircase etc., is also the double pole switch of special kind.



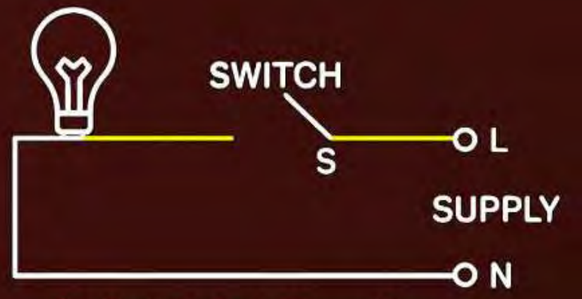
SWITCHES

Safety precaution while using a switch

A switch should never be touched with wet hands. If water reaches the live wire, it forms a conducting layer between the hand and the live wire of the switch due to which a current passes to the hand through it and the person may get a fatal shock.



(a) Switch in 'on' position



(b) Switch in 'off position

CIRCUITS WITH DUAL CONTROL SWITCHES (STAIRCASE WIRING)

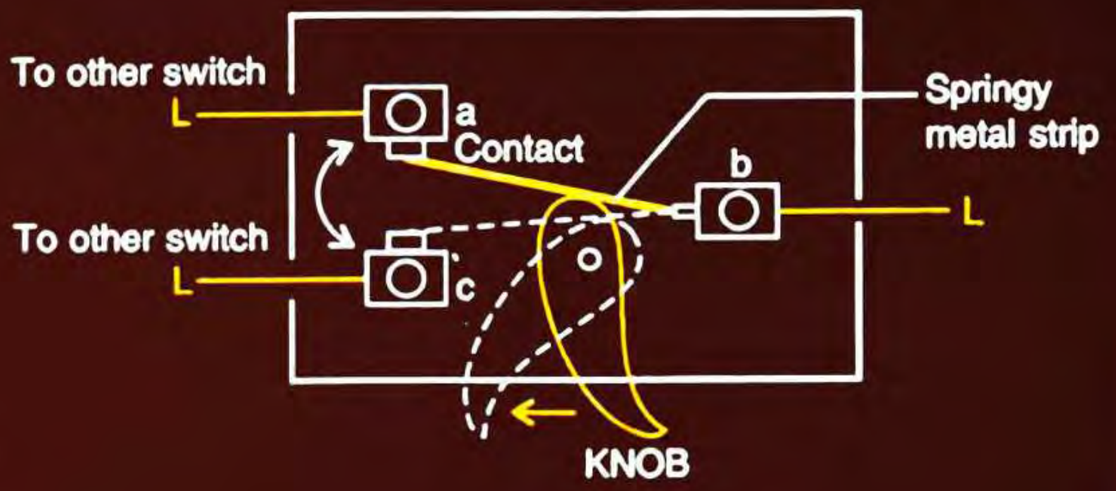
A dual control switch is a special double-pole switch used to control an appliance, such as a bulb, from two different places.

It is commonly used at both ends of a staircase or a long corridor.

The switch has two possible connections:

- Position 1: Terminal b connects to a → **connection ba.**
- Position 2: Terminal b connects to c → **connection bc.**

Thus, the bulb can be switched ON or OFF from either location.



CIRCUITS WITH DUAL CONTROL SWITCHES (STAIRCASE WIRING)

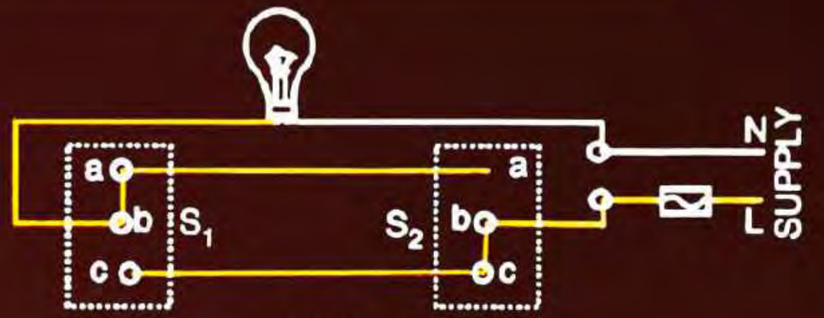
Two switches S_1 and S_2 are used—one at the bottom and one at the top of the staircase.

The bulb can be switched ON or OFF from either switch independently.

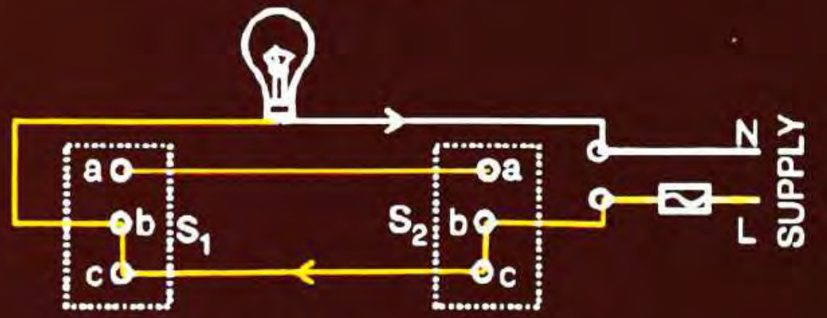
- When S_1 is operated, connection ba → bc, completing the circuit and turning the bulb ON.
- Similarly, operating S_2 changes bc → ba, completing the circuit and turning the bulb ON.

If the bulb is ON, operating either S_1 or S_2 changes the connection and breaks the circuit, turning the bulb OFF.

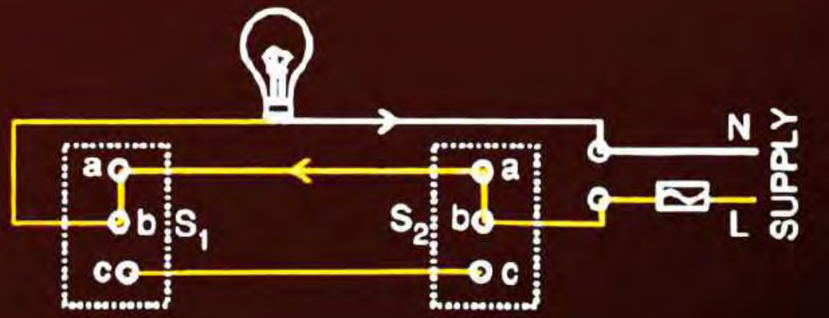
Key point: Either switch can independently control the bulb without depending on the position of the other switch.



(a) Bulb off



(b) Bulb on through switch S_1



(c) Bulb on through switch S_2



QUESTIONS

A switch must be connected in the:

1. live wire
2. neutral wire
3. earth wire
4. either earth or neutral wire.



QUESTIONS

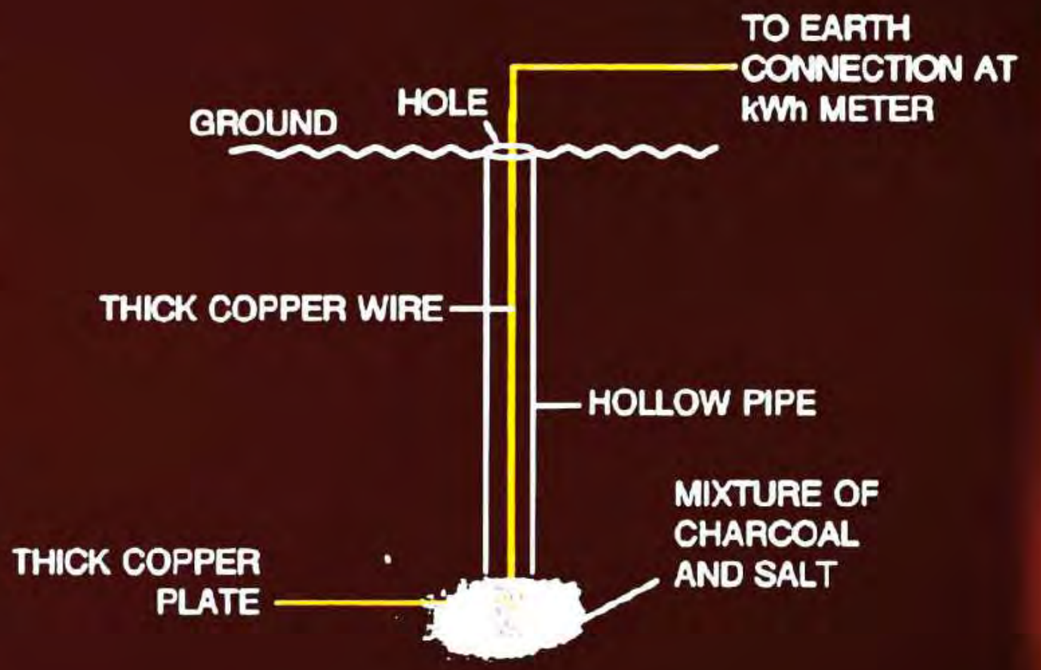
A main switch in the main distribution board is present in

1. A live wire
2. a neutral wire
3. a live as well as neutral wire
4. an earth wire

EARTHING (GROUNDING)

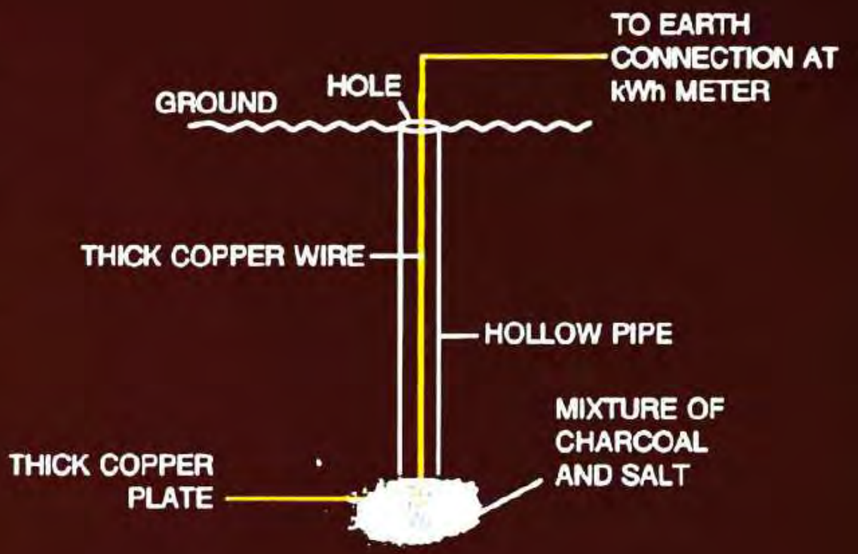
(a) Local earthing

- Local earthing is done near the kWh meter of a house.
- A hole about 2–3 m deep is dug in the ground.
- A **copper rod/thick copper wire** is inserted into the hole.
- A 50 cm × 50 cm copper plate is attached to the lower end and buried in the ground.
- **The plate is surrounded by charcoal and salt to provide good electrical contact with the earth.**
- **Key Point**: Earthing provides a safe path for fault current to flow into the ground, protecting people and appliances.



EARTHING (GROUNDING)

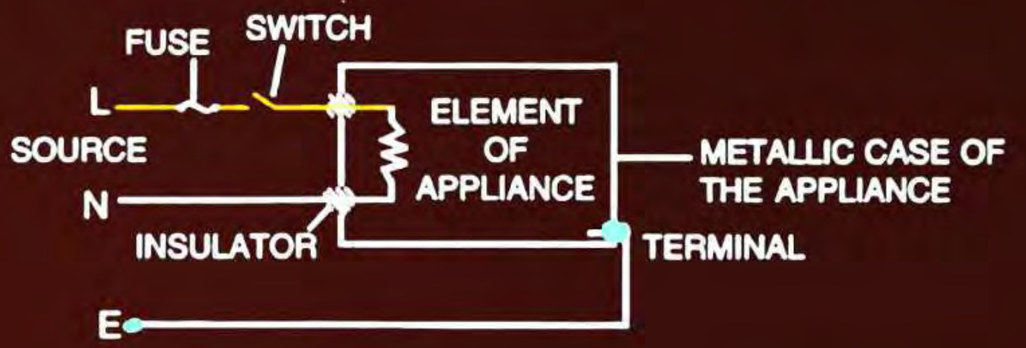
- Water is poured through the pipe from time to time to keep the ground damp and improve conductivity.
- This creates a conducting path between the copper plate and the earth.
- The upper end of the copper rod is connected to the earth connection at the kWh meter.
- Safety: During a short circuit or excessive current, the fault current flows safely into the earth through the earth wire.
- This prevents the line wires from overheating, reducing the risk of fire and electrical damage.



Earthing of an Appliance

- It is essential to provide a connection for earthing an electrical appliance having a metallic case (such as refrigerator, toaster, geyser, electric iron, electric cooler, etc.) which we handle physically.
- For this, the **earth wire of the cable is connected to the terminal provided on the outer metallic case of the appliance**.

- The switch and the fuse are connected in the live wire. The earth wire is connected to the metallic case of appliance.



Earthing of an Appliance



- The earthing symbol is used to show the earth connection of an appliance.
- The paint on a metallic case acts as an insulator, so it must be removed at the point where the earth connection is made.
- If the live wire touches the metallic case due to damaged insulation, the case can become highly charged and cause a dangerous electric shock.
- With proper earthing, the fault current gets a low-resistance path to the earth through the metallic case.
- The resulting heavy current causes the fuse to blow, disconnecting the appliance from the supply.
- Thus, earthing protects both the person from electric shock and the appliance from damage.

THREE-PIN PLUG

- A three-pin plug connects an electrical appliance to the mains supply and has three brass pins:
  - **E → Earth (top)**
  - **L → Live (left)**
  - **N → Neutral (right)**
- The pins are split at their ends to provide a tight connection.
- The earth pin is thicker and longer than the other two. Its longer length ensures that the earth connection is made first, improving safety.
- Its greater thickness prevents it from being accidentally inserted into the live or neutral socket.
- If an appliance is faulty, the earth connection provides a path for fault current, causing the fuse to blow and disconnect the supply.





QUESTIONS

In a three pin plug the live pin in :

1. thinner and is toward left
2. thicker and is towards right
3. thicker and is towards left
4. thinner and is towards right

Socket

- A socket is a fixture in an electric circuit into which the plug is inserted.
- It has three holes with hollow metallic tubes, usually made of brass, connected to the supply wires.
- **Hole connections:**
  - **Upper, bigger hole → Earth**
  - **Right hole → Live**
  - **Left hole → Neutral**
- **Safety precautions:**
- Always keep your hands completely dry while inserting a plug.
- Ensure the plug fits tightly in the socket.
- A loose connection can cause sparking and overheating, which may damage or burn the plug/socket.



COLOUR CODING OF WIRES IN A CABLE

Wire	Colour	
	Old Convention	New Convention
Live	Red	Brown
Neutral	Black	Light Blue
Earth	Green	Green or Yellow

HIGH TENSION WIRES

- Every wire can withstand only a specific maximum current. Excess current can cause overheating and fire.
- For high voltage and heavy current, special high tension wires are used.
- A high tension wire has low resistance and a large surface area.
- It is made by twisting several thin, insulated wires together instead of using one thick wire.
- The large surface area helps the wire radiate heat more effectively to the surroundings.

$R \propto \frac{1}{A}$







SAFETY PRECAUTIONS WHILE USING THE ELECTRICITY

🔥 **Prevention of Fire**

- Fire can occur due to overheating of wires, caused by short circuiting or damaged insulation.
- Use wires with a current-carrying capacity greater than the maximum current that can flow when all appliances operate together.

⚡ **Prevention of Electric Shock**

- Electric shock may occur due to poor insulation, wet hands, lack of earthing, absence of a fuse, or improper local earthing.

Five important precautions:

- Use good-quality wire insulation and check it regularly.
- Never touch or operate appliances with wet hands.
- Properly earth the metallic case of every appliance.
- Provide a proper fuse in the live wire of each appliance.
- Ensure proper local earthing near the kWh meter.

SUPERCONDUCTORS

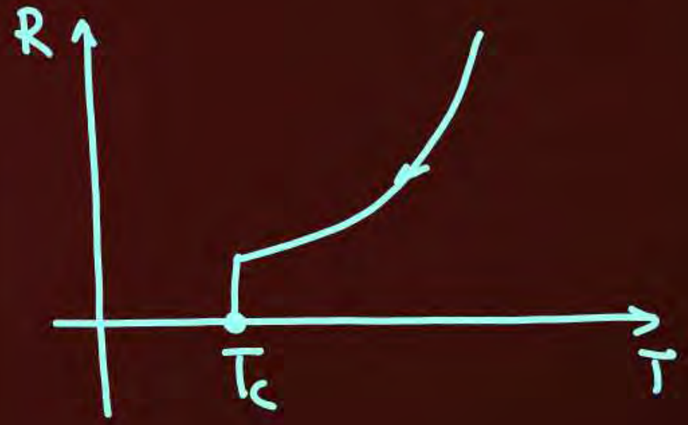
- In 1911, H. Kammerling Onnes discovered that the resistance of mercury becomes zero at 4.2 K.
- The phenomenon in which a substance shows almost zero electrical resistance at very low temperatures is called superconductivity.
- Materials that exhibit this phenomenon are called superconductors.
- The temperature at which a substance changes into a superconductor is called its critical temperature (Tc) or transition temperature.

Critical Temperature of Some Superconductors

$0^\circ C \Rightarrow 273\ K$

$-270^\circ C \Rightarrow 3\ K$

Substance	Tc
Mercury	4.2 K
Zinc	0.79 K
Lead	7.26 K
Vanadium	4.3 K
Niobium	9.22 K



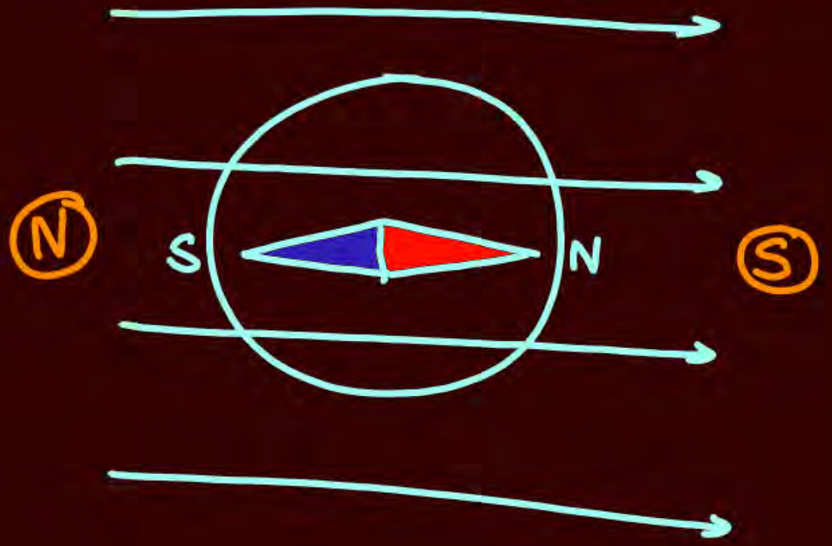


OERSTED'S EXPERIMENT ON THE MAGNETIC EFFECT OF ELECTRIC CURRENT

Magnetic Compass or Needle

- Hans Oersted (1820) discovered that an electric current produces a magnetic field around a conducting wire.
- The presence of this magnetic field can be detected using a compass needle.
- In the absence of a magnetic field, a compass needle can point in any direction.
- In a magnetic field, the needle aligns itself along the direction of the magnetic field.
- Due to Earth's magnetic field, a compass needle normally points in the North–South direction


N
S


N
S
S
N


N
S



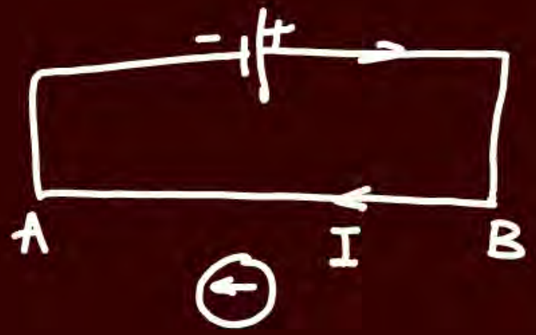








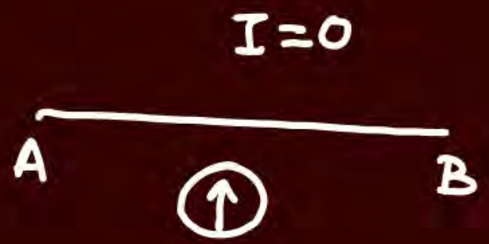
$$B \propto \frac{1}{r}$$









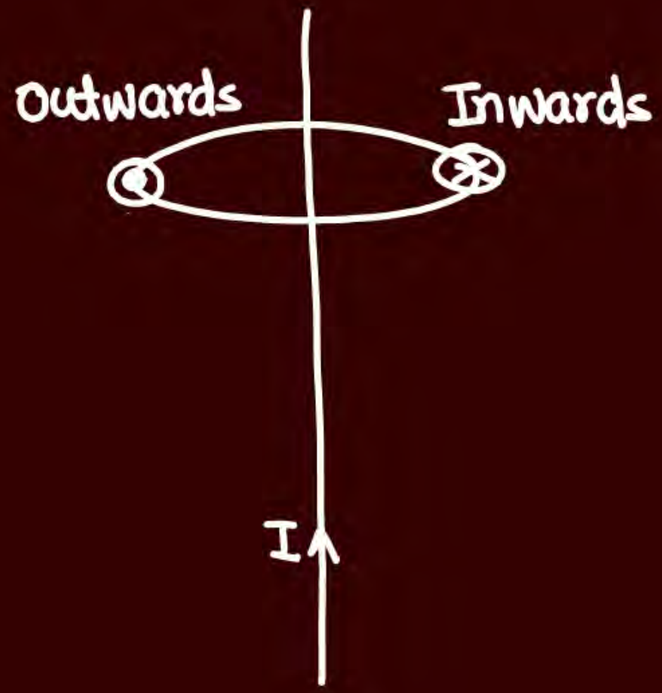


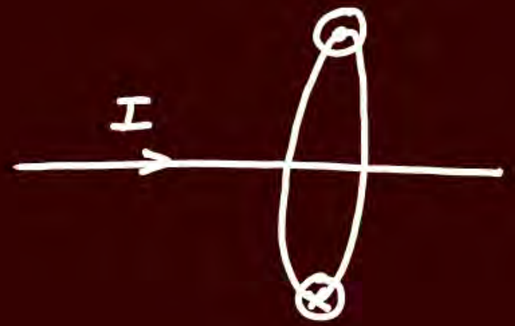


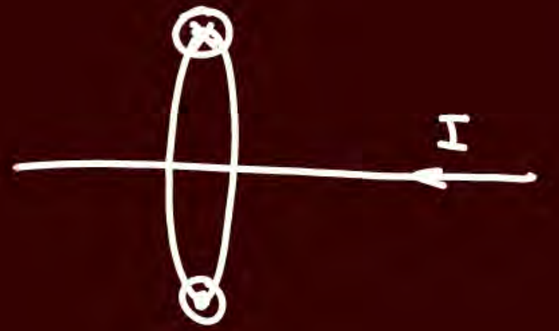




Direction of MF





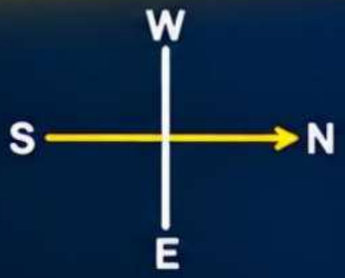


⊙ Outwards

⊗ Inwards.

OERSTED'S EXPERIMENT





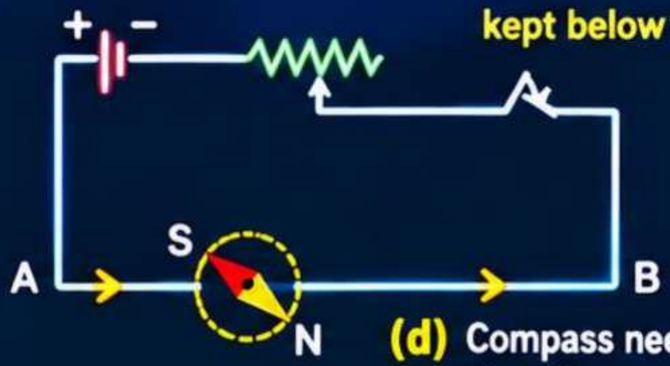
(a) Key open and compass needle kept below the wire



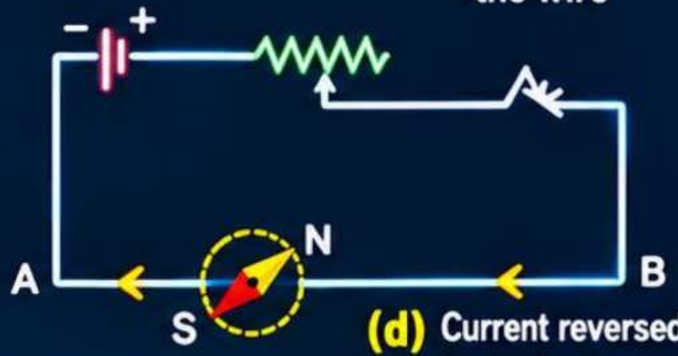
(b) Key closed and compass needle kept below the wire



(c) Current reversed and compass needle kept below the wire



(d) Compass needle placed above the wire



(d) Current reversed and compass needle kept above the wire



OERSTED'S EXPERIMENT ON THE MAGNETIC EFFECT OF ELECTRIC CURRENT

- A current-carrying straight conductor behaves like a magnet.
- The magnetic effect increases when the current increases.

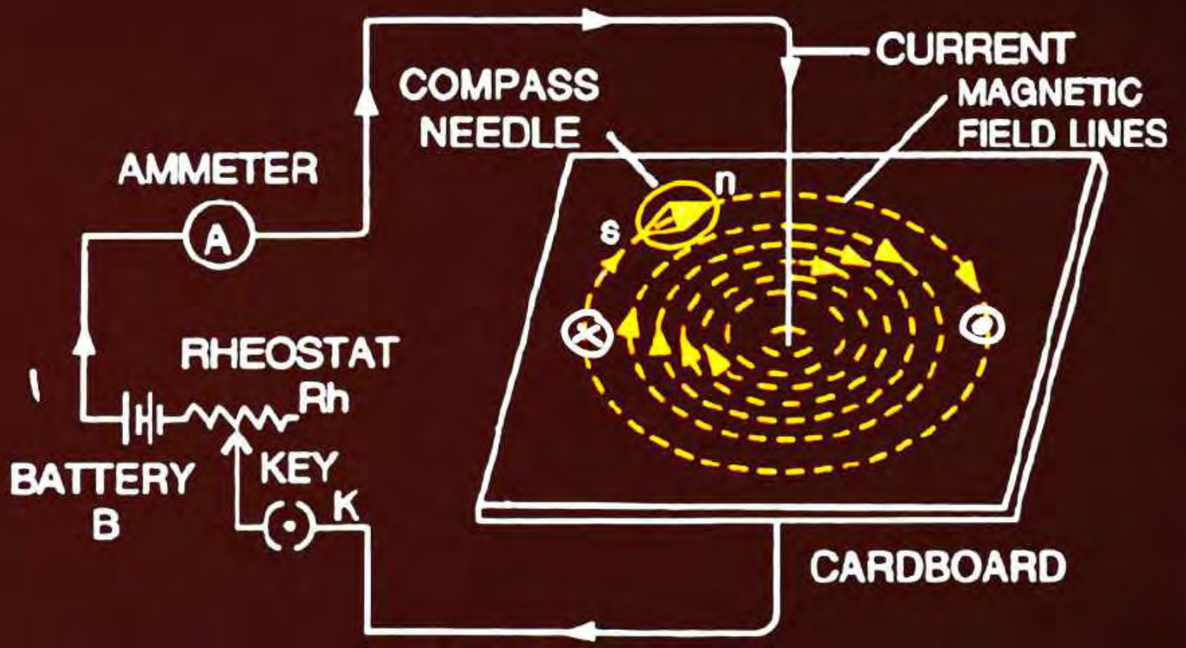
- The magnetic field around the conductor is perpendicular (at right angles) to the direction of current.

- The direction of deflection of the compass needle depends on:
- The direction of current in the conductor.

- The position of the conductor relative to the compass needle—whether it is above or below the needle.

MAGNETIC FIELD AND FIELD LINES DUE TO CURRENT IN A STRAIGHT WIRE

- Magnetic field lines around a straight current-carrying wire are concentric circles with their centres on the wire and their planes perpendicular to it.
- Reversing the current reverses the direction of the magnetic field, although the circular pattern of field lines remains unchanged.
- Increasing the current makes the magnetic field lines denser and extends the magnetic effect farther from the wire, showing that magnetic field strength increases.





MAGNETIC FIELD AND FIELD LINES DUE TO CURRENT IN A STRAIGHT WIRE

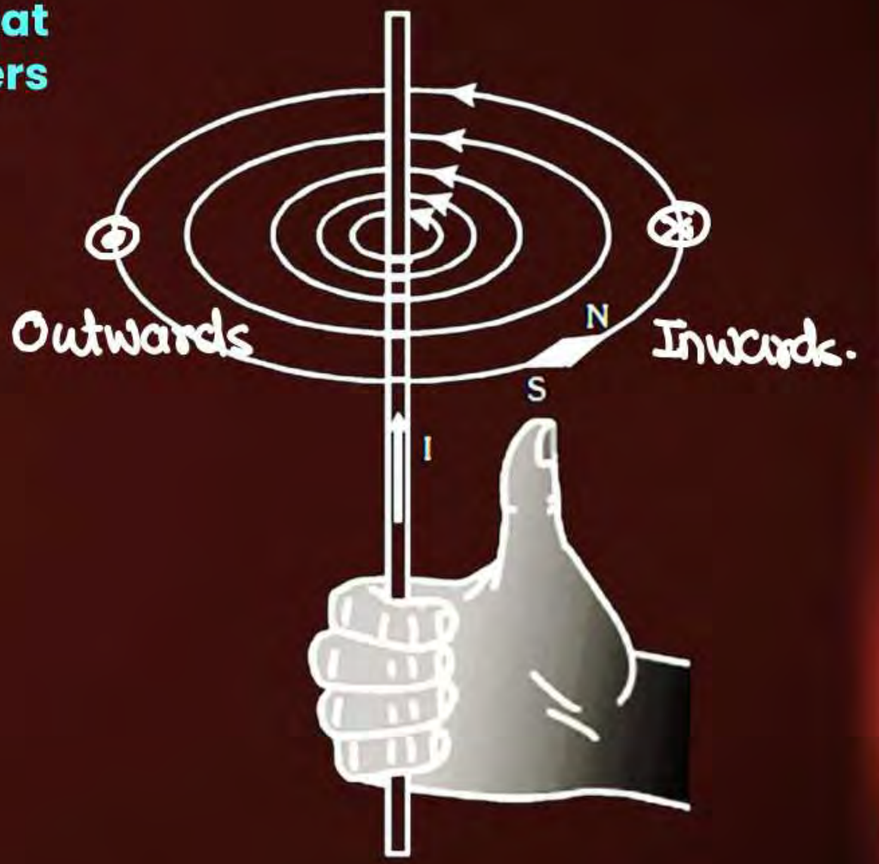
- The magnetic field at any point is the resultant of two fields:
  - Magnetic field due to the current-carrying wire
  - Earth's magnetic field
- Near the wire: The wire's magnetic field is stronger, so iron filings form circles.
-
- Far from the wire: Earth's magnetic field becomes dominant, so iron filings tend to form straight lines.

- A neutral point is a point where the two magnetic fields are equal in magnitude and opposite in direction.

- At a neutral point, the resultant magnetic field is zero, so the compass needle can rest in any direction.

RIGHT HAND THUMB RULE

If we hold the current carrying conductor in our right hand such that the thumb points in the direction of flow of current, then the fingers encircle the wire in the direction of the magnetic field lines.



QUESTIONS

The presence of magnetic field at a point can be detected by means of:

1. a strong magnet
2. a solenoid
3. a compass needle
4. a current carrying wire



QUESTIONS

On reversing the direction of current in a wire, the magnetic field produced by it:

1. gets reversed in direction
2. increases in strength
3. decreases in strength
4. remains unchanged in strength and direction



QUESTIONS

Why does a magnetic needle show a deflection when brought close to a current carrying conductor?

A current carrying conductor produces a magnetic field around it and the magnetic needle in this magnetic field experiences a torque due to which it deflects to align itself in the direction of magnetic field.

HOMEWORK

- REVISE THIS LECTURE AGAIN.
- PRACTISE 20 QUESTIONS DAILY.
- COMMENT YOUR RESPONSE ABOUT THIS CLASS.

Thank You